Problems for Practice

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Practice Sheet - DFA

1. Draw a DFA for the set of binary strings that start with **01**. ∑ = {0,1}
2. Draw a DFA for the set of binary strings that are divisible by 8 while considered as binary numbers. ∑ = {0,1}
3. Draw a DFA for the set of strings that end with **abb**. ∑= {a, b, c}
4. Draw a DFA for the set of binary strings that have an even number of **0**'s or an odd number of **1**'s. ∑ = {0,1}
5. Draw a DFA for the set of strings that have **011** as a substring and **001** as not a substring. ∑ = {0,1}
6. Draw a DFA for the set of strings that have a length of at least **4**. ∑ = {a, b}
7. Draw a DFA for the set of binary strings that contain at least three **1**'s. ∑ = {0,1}
8. Draw a DFA for the set of strings that have exactly three **a**'s. ∑= {a, b, c}
9. Draw a DFA for the set of strings that have lengths of not more than **6**. ∑= {0,1}
10. Draw a DFA for the set of strings that have exactly three **1**'s and four **0**'s. ∑= {0,1,2}
11. Draw a DFA for the set of strings that have **three** consecutive **1**'s. ∑= {0,1}

SOME MORE PROBLEMS WITH VIDEO SOLUTIONS

1. DFA of strings which contains substring "aba" and has b in the second last position
https://youtu.be/NnLUV8ele4Q
2. DFA of strings that have 0 as every 3rd symbol
https://youtu.be/XLAYQvl_yMM
3. DFA of Set of all strings that are at least of length 4 and contain an even number of 1's
https://youtu.be/fl6PAFTXIl4

4. DFA of strings where the 3rd last symbol is 0
https://youtu.be/_eOp_uRvDYg
5. Draw a DFA for the set of binary strings that start with 01. $\sum = \{0,1\}$
https://youtu.be/0T8q6MfJQv8
6. Draw a DFA for the set of strings that end with abb. $\sum= \{a, b, c\}$
https://youtu.be/c7qGrkPhoZ4
7. DFA for the set of binary strings that have an even number of 0's or an odd number of 1's. $\sum = \{0,1\}$
https://youtu.be/Y4pIThlKqq8
8. Draw a DFA for the set of strings that have a length of at least 4. $\sum = \{a, b\}$
https://youtu.be/A7Fylo0VWV8
9. Draw a DFA for the set of binary strings that contain at least three 1's. $\sum = \{0,1\}$
https://youtu.be/cUaYXNGOFGk
10. Draw a DFA for the set of strings that have exactly three a's. $\sum= \{a, b, c\}$
https://youtu.be/ee-ufKPAf9w
11. Draw a DFA for the set of strings that have lengths of not more than 6. $\sum= \{0,1\}$
https://youtu.be/gs68vLOZ0w0
12. Draw a DFA for the set of strings that have three consecutive 1's. $\sum= \{0,1\}$
https://youtu.be/YYHnEzOASeE